🐾 SMART PET FEEDER & WATER DISPENSER (ESP32 + Telegram)

🔧 1. Project Overview

Sistem ini direka untuk memberi makanan dan air kepada haiwan peliharaan secara automatik. Ia menggunakan **ESP32**, beberapa **sensor paras**, **servo motor**, dan **Telegram Bot** supaya pengguna boleh:

- Tetapkan masa pemberian makanan
- Pantau paras makanan & air
- Terima notifikasi bila bekalan hampir habis

🧩 2. Component List

No	Component	Quantity
1	ESP32 Dev Board	1
2	Ultrasonic Sensor (HC-SR04)	2
3	Non-contact Water Level Sensor	1
4	Water Level Sensor Module (Height Surface Type)	1
5	SG90 Servo Motor	2
6	Jumper Wires	-
7	Power Supply 5V 2A	1
8	Resistors (330Ω optional)	-
9	Breadboard	1

⚙️ 3. Wiring Connection

Component	Pin Label	Connection To (ESP32 Pin)	Description / Function
Ultrasonic Sensor (Food Storage)	VCC	5V	Bekalan kuasa sensor
	GND	GND	Ground sambungan bersama
	TRIG	GPIO14	Hantar signal trigger bunyi ultrasonik
	ECHO	GPIO27	Terima signal echo untuk ukur jarak makanan

Component	Pin Label	Connection To (ESP32 Pin)	Description / Function
Ultrasonic Sensor (Food Bowl)	VCC	5V	Bekalan kuasa sensor
	GND	GND	Ground sambungan bersama
	TRIG	GPIO26	Hantar signal trigger ultrasonik
	ECHO	GPIO25	Terima signal echo untuk ukur paras makanan di mangkuk
Non-Contact Water Level Sensor (Water Tank)	VCC	5V	Bekalan kuasa sensor
	GND	GND	Ground sambungan bersama
	D0	GPIO33	Output digital (HIGH bila ada air, LOW bila air kurang)
Water Level Sensor Module (Water Bowl)	VCC	5V	Bekalan kuasa sensor
	GND	GND	Ground sambungan bersama
	A0	GPIO34 (Analog Input)	Bacaan analog paras air di bekas minum
Servo Motor (Food Gate)	VCC (Red)	5V	Bekalan kuasa motor servo
	GND (Brown)	GND	Ground sambungan bersama
	Signal (Orange)	GPIO13	Kawal bukaan pintu makanan
Servo Motor (Water Valve)	VCC (Red)	5V	Bekalan kuasa motor servo
	GND (Brown)	GND	Ground sambungan bersama
	Signal (Orange)	GPIO12	Kawal injap buka/tutup air
ESP32 Power Supply	VIN	5V (USB/Adapter)	Sumber kuasa utama
	GND	GND	Ground rujukan keseluruhan sistem

⚠️ Wiring Notes

- Semua **GND mesti bersambung bersama (common ground)**.
- Gunakan **power supply 5V 2A** untuk stabilkan servo.
- Jika servo bergetar atau restart ESP32, tambah **capacitor 470µF – 1000µF** antara 5V dan GND berhampiran servo.
- Untuk sensor analog (GPIO34), pastikan **tidak ada input voltage > 3.3V** (gunakan divider resistor jika perlu).

- Non-contact sensor diletakkan di **luar dinding tangki air**, tidak menyentuh air.
- Ultrasonic sensor perlu menghadap ke bawah ke arah makanan/air.

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5. Full Arduino IDE Code

```
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <Servo.h>

#define WIFI_SSID "YourWiFiName"
#define WIFI_PASSWORD "YourWiFiPassword"
#define BOT_TOKEN "1234567890:ABCDefGhiJKlmNoPQRstuVWxyz"
#define CHAT_ID "123456789"

WiFiClientSecure client;
UniversalTelegramBot bot(BOT_TOKEN, client);

// ====== Sensor Pins ======
#define TRIG_FOOD_STORAGE 14
#define ECHO_FOOD_STORAGE 27
#define TRIG_FOOD_BOWL 26
#define ECHO_FOOD_BOWL 25
#define WATER_STORAGE_SENSOR 33
#define WATER_LEVEL_SENSOR 34

// ====== Servo Pins ======
#define SERVO_FOOD 13
#define SERVO_WATER 12

Servo servoFood;
Servo servoWater;

// ====== Timing ======
unsigned long lastCheckTime = 0;
unsigned long interval = 2000;

// ====== Helper Function (Ultrasonic) ======
long readUltrasonic(int trigPin, int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH);
  long distance = duration * 0.034 / 2;
  return distance;
}

// ====== Setup ======
void setup() {
  Serial.begin(115200);
```

```
  pinMode(TRIG_FOOD_STORAGE, OUTPUT);
  pinMode(ECHO_FOOD_STORAGE, INPUT);
  pinMode(TRIG_FOOD_BOWL, OUTPUT);
  pinMode(ECHO_FOOD_BOWL, INPUT);
  pinMode(WATER_STORAGE_SENSOR, INPUT);
  pinMode(WATER_LEVEL_SENSOR, INPUT);

  servoFood.attach(SERVO_FOOD);
  servoWater.attach(SERVO_WATER);

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  client.setInsecure();
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected!");
  bot.sendMessage(CHAT_ID, "🐾 Smart Pet Feeder System is Online!", "");
}

// ====== Main Loop ======
void loop() {
  if (millis() - lastCheckTime > interval) {
    lastCheckTime = millis();

    // Baca sensor makanan
    int foodStorage = readUltrasonic(TRIG_FOOD_STORAGE, ECHO_FOOD_STORAGE);
    int foodBowl = readUltrasonic(TRIG_FOOD_BOWL, ECHO_FOOD_BOWL);

    // Baca sensor air
    int waterStorage = digitalRead(WATER_STORAGE_SENSOR);
    int waterLevel = analogRead(WATER_LEVEL_SENSOR);

    Serial.printf("FoodStorage:%dcm FoodBowl:%dcm WaterLevel:%d\n",
foodStorage, foodBowl, waterLevel);

    // Logik Makanan
    if (foodBowl > 10) { // Bila makanan kurang
      servoFood.write(90);
      delay(1000);
      servoFood.write(0);
      bot.sendMessage(CHAT_ID, "🍖 Food dispensed automatically!", "");
    }

    // Logik Air
    if (waterLevel < 500) { // Paras air rendah
      servoWater.write(90);
      delay(2000);
      servoWater.write(0);
      bot.sendMessage(CHAT_ID, "💧 Water refilled automatically!", "");
    }

    // Amaran
    if (foodStorage > 15) {
      bot.sendMessage(CHAT_ID, "⚠️ Food storage is running low!", "");
```

```
    }
    if (waterStorage == LOW) {
      bot.sendMessage(CHAT_ID, "⚠️ Water tank is low! Please refill.", "");
    }
  }

  // Check command Telegram
  int numNewMessages = bot.getUpdates(bot.last_message_received + 1);
  while (numNewMessages) {
    Serial.println("New message from Telegram");
    for (int i = 0; i < numNewMessages; i++) {
      String text = bot.messages[i].text;
      if (text == "/status") {
        String msg = "📊 *Status:*\n";
        msg += "Food Storage: " + String(readUltrasonic(TRIG_FOOD_STORAGE,
ECHO_FOOD_STORAGE)) + " cm\n";
        msg += "Food Bowl: " + String(readUltrasonic(TRIG_FOOD_BOWL,
ECHO_FOOD_BOWL)) + " cm\n";
        msg += "Water Tank: " + String(digitalRead(WATER_STORAGE_SENSOR)) +
"\n";
        msg += "Water Level: " + String(analogRead(WATER_LEVEL_SENSOR)) +
"\n";
        bot.sendMessage(CHAT_ID, msg, "Markdown");
      }
    }
    numNewMessages = bot.getUpdates(bot.last_message_received + 1);
  }
}
```

🧠 6. System Flow

1. ESP32 akan baca sensor setiap 2 saat.
2. Jika makanan di mangkuk rendah → servo buka pintu makanan.
3. Jika air di mangkuk rendah → servo buka injap air.
4. Jika bekalan air atau makanan dalam tangki simpanan rendah → Telegram beri amaran.
5. Pengguna boleh taip `/status` untuk semak paras semasa.